

The j -formula for the family $\lambda = (3, 1^{n-3})$ and a Catalan support phenomenon for hook partitions

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Abstract

We extend the May-11 structural proof of the j -formula from $\lambda = (2, 1^{n-2})$ to the family $\lambda = (3, 1^{n-3})$ at every $n \geq 6$. The proof tracks the image of the staircase product $R'_{n-1}R'_n$ on V_λ step-by-step in the seminormal basis. After the first staircase factor R'_n , the image has dimension $n - 2$ and admits an explicit basis of $+q$ -eigenvectors of T_{n-1} . After the second staircase factor R'_{n-1} , the image collapses to dimension 1 supported on exactly 5 standard tableaux, all in E_1^- . As a corollary, $\Pi^{S_n}|_{V_{(3,1^{n-3})}} = 0$ unconditionally for every $n \geq 5$.

We further document a striking computational phenomenon: for any hook $\lambda = (k, 1^{n-k})$ in the rank-zero regime, the unique (up to scalar) spanning vector of $B_{\ell+1}V_\lambda$ has support of size exactly C_k (the k -th Catalan number) on subsets of $\{n - 2k + 3, \dots, n\}$. Verified for $k \leq 4$ at $n \leq 8$. This extends the rank-corner formula and suggests a Catalan structure governing the kernel of the staircase Bott–Samelson element on hook irreducibles.

1 Setup

Notation as in `2026-05-11-j-formula.tex`. $H_q(S_n)$ has generators $T_1, \dots, T_{n-1}$. V_λ is the seminormal cell module of $H_q(S_n)$ for the partition $\lambda \vdash n$, with basis $\{v_T\}$ indexed by SYTs T of shape λ .

The staircase Bott–Samelson element is

$$\Pi^{S_n} = R'_2 R'_3 \cdots R'_n, \quad R'_k = (T_{k-1} + 1)(T_{k-2} + 1) \cdots (T_1 + 1).$$

Write $B_j := R'_j R'_{j+1} \cdots R'_n$. $E_i^\pm$ denotes the $\pm$ -eigenspace of T_i (eigenvalues q resp. -1).

1.1 The seminormal basis of a hook

For $\lambda = (k, 1^{n-k})$, an SYT T is determined by the $(k - 1)$ -element subset $S \subset \{2, \dots, n\}$ that fills row 1 (after the entry 1 at $(1, 1)$). We write $v_S := v_T$. The values $\{2, \dots, n\} \setminus S$ fill column 1 in cells $(2, 1), (3, 1), \dots, (n - k + 1, 1)$ in increasing order.

Lemma 1 (T_i -action on hook seminormal basis). *For $i \in \{1, \dots, n - 1\}$ and SYT T of shape $(k, 1^{n-k})$ indexed by $S \subset \{2, \dots, n\}$:*

- If $i, i+1 \in \{1\} \cup S$ (both in row 1): $T_i v_S = q v_S$.
- If $i, i+1 \notin \{1\} \cup S$ (both in column 1): $T_i v_S = -v_S$.
- Otherwise (exactly one of $i, i+1$ in S): T_i acts as a 2×2 Heffsmit block on $\text{span}(v_S, v_{s_i S})$ where $s_i S$ swaps i and $i+1$ in S (or, equivalently, in row 1 versus column 1).

Proof. Standard. In the seminormal basis, $T_i v_T$ depends on the relative positions of i and $i+1$ in T : same row $\Rightarrow q$, same column $\Rightarrow -1$, otherwise T_i couples T with $s_i T$ (the SYT obtained by swapping i and $i+1$). In the hook, i and $i+1$ are in row 1 iff both lie in $\{1\} \cup S$, and in column 1 iff neither does. $\square$

Remark 2 (E_1^- for hooks). $T_1 v_S = q v_S$ if $2 \in S$, and $T_1 v_S = -v_S$ otherwise. Hence $E_1^- = \text{span}\{v_S : 2 \notin S\}$, the SYTs with 2 in column 1.

1.2 Hoefsmit $+q$ -eigenvectors

When T_i acts as a 2×2 block on $\text{span}(v_S, v_{s_i S})$, the eigenvalues are q and -1 . The $+q$ -eigenvector is unique up to scalar:

$$u_{S, s_i S}^+ = \alpha_d v_S + \beta_d v_{s_i S},$$

where d is the axial distance from the cell of i to the cell of $i+1$ in T (content of the latter minus content of the former, with our sign convention; the explicit formulas for α_d, β_d will not matter, only that they are nonzero generically in q). The operator (T_i+1) acts on $\text{span}(v_S, v_{s_i S})$ by projecting onto the $+q$ -eigenspace, scaled by $1+q$:

$$(T_i+1) v_S = c_d u_{S, s_i S}^+, \quad (T_i+1) v_{s_i S} = c'_d u_{S, s_i S}^+,$$

with $c_d, c'_d \in \mathbb{Q}(q)$ generically nonzero.

2 Theorem and main proof

Theorem 3 (j -formula for $\lambda = (3, 1^{n-3})$). *For every $n \geq 6$, $\lambda = (3, 1^{n-3}) \vdash n$ has length $\ell = n - 2$ and the j -formula holds with $j = \ell + 1 = n - 1$:*

$$B_{n-1}(V_{(3, 1^{n-3})}) := R'_{n-1} R'_n V_{(3, 1^{n-3})} \subseteq E_1^-.$$

Moreover, the image is one-dimensional, supported on exactly 5 SYTs indexed by the subsets

$$\{(n-3, n-2), (n-3, n-1), (n-2, n-1), (n-2, n), (n-1, n)\}.$$

Corollary 4. $\Pi^{S_n}|_{V_{(3, 1^{n-3})}} = 0$ unconditionally for every $n \geq 6$.

Proof. For $n \geq 6$, $\lambda = (3, 1^{n-3})$ is in the rank-zero regime ($\ell = n - 2 > \lceil n/2 \rceil$ iff $n \geq 5$, and the boundary $n = 5$ is excluded since $\ell = 3 = \lceil 5/2 \rceil$ is not strict). By Theorem 3, $B_{n-1} V_\lambda \subseteq E_1^-$. The next factor of $\Pi^{S_n} = R'_2 \cdots R'_n$ to act after B_{n-1} is $R'_{n-2} = (T_{n-3}+1) \cdots (T_1+1)$, whose rightmost factor (T_1+1) annihilates E_1^- . Hence $\Pi^{S_n} V_\lambda = 0$. (For $n = 5$, $V_{(3, 1, 1)}$ is at the rank-zero threshold, not in the regime, and indeed $\Pi^{S_5}|_{V_{(3, 1, 1)}}$ has rank 1, as proved in May-7-8.) $\square$

Proof of Theorem 3. The proof is by step-by-step computation in the seminormal basis, tracking the image after each Hecke-generator factor of $R'_{n-1} R'_n$. We split into two phases: tracking R'_n (Phase A, Section 3), then tracking R'_{n-1} on the result of Phase A (Phase B, Section 4). $\square$

3 Phase A: the image of R'_n

Let $W_j^{(n)}$ denote the image of the partial product $(T_j+1)(T_{j-1}+1)\cdots(T_1+1)$ applied to $V_{(3,1^{n-3})}$, for $1 \leq j \leq n-1$.

Proposition 5 (Basis of $W_j^{(n)}$). *For $2 \leq j \leq n-1$,*

$$W_j^{(n)} = \text{span} \left(\underbrace{\{u_{(c,j)}^+ : 2 \leq c \leq j-1\}}_{j-2 \text{ vectors}} \cup \{v_{(j,j+1)}\} \cup \underbrace{\{u_{(j,b)}^+ : j+2 \leq b \leq n\}}_{n-j-1 \text{ vectors}} \right), \quad (1)$$

where:

- $u_{(c,j)}^+ = \alpha_j v_{(c,j)} + \beta_j v_{(c,j+1)}$ is the $+q$ -eigenvector of T_j in the 2-block $\{v_{(c,j)}, v_{(c,j+1)}\}$, acting at axial distance $d = j$ (computed below);
- $u_{(j,b)}^+ = \alpha_j v_{(j,b)} + \beta_j v_{(j+1,b)}$ is the $+q$ -eigenvector of T_j in the 2-block $\{v_{(j,b)}, v_{(j+1,b)}\}$, acting at the same axial distance $d = j$.

The dimension of $W_j^{(n)}$ is $(j-2) + 1 + (n-j-1) = n-2$.

Proof. By induction on j , with base case $j=1$ handled separately.

Base case $j=1$. $T_1 v_S = q v_S$ if $2 \in S$, else $-v_S$, so (T_1+1) projects to $E_1^+ = \text{span}\{v_S : 2 \in S\}$ (scaled by $1+q$). For $\lambda = (3, 1^{n-3})$, this is $\text{span}\{v_{(2,b)} : 3 \leq b \leq n\}$, dimension $n-2$.

Step $j=2$. Apply (T_2+1) to $W_1^{(n)}$. For each $v_{(2,b)}$, locate 2 and 3 in $T_{(2,b)}$:

- $b=3$: both 2 and 3 are in row 1 (positions $(1,2), (1,3)$). $T_2 v_{(2,3)} = q v_{(2,3)}$, so $(T_2+1)v_{(2,3)} = (1+q)v_{(2,3)}$.
- $b \geq 4$: 2 at $(1,2)$ (content 1), 3 at $(2,1)$ (content -1). T_2 couples $v_{(2,b)}$ with $v_{(3,b)}$ (swap of $2,3$), axial distance $d = 1 - (-1) = 2$.

By Lemma 1 and the $+q$ -projection formula,

$$(T_2+1)v_{(2,b)} = c_2 u_{(2,b)}^+, \quad u_{(2,b)}^+ = \alpha_2 v_{(2,b)} + \beta_2 v_{(3,b)}.$$

Hence $W_2^{(n)} = \text{span}\{v_{(2,3)}\} \cup \{u_{(2,b)}^+ : 4 \leq b \leq n\}$. Compare with (1): the formula gives $\{u_{(c,2)}^+ : 2 \leq c \leq 1\} \cup \{v_{(2,3)}\} \cup \{u_{(2,b)}^+ : 4 \leq b \leq n\}$, where the first set is empty. Match.

Inductive step. Fix $j \geq 3$ and assume (1) holds for $j-1$. We compute (T_j+1) on each basis vector of $W_{j-1}^{(n)}$.

(i) $u_{(c,j-1)}^+ = \alpha_{j-1} v_{(c,j-1)} + \beta_{j-1} v_{(c,j)}$ for $2 \leq c \leq j-2$. By Lemma 1:

- In $v_{(c,j-1)}$ (where $S = \{c, j-1\}$, $c < j-1$): $j \notin S$, so j is in column 1. Its position: column 1 contains $\{2, \dots, n\} \setminus \{c, j-1\}$ in increasing order; the elements below j are $\{2, \dots, j-1\} \setminus \{c, j-1\} = \{2, \dots, j-2\} \setminus \{c\}$ ($j-3$ elements). So j is at $(j-2, 1)$, content $-(j-3)$. Similarly $j+1$ is at $(j-1, 1)$, content $-(j-2)$. Same column, adjacent. $T_j v_{(c,j-1)} = -v_{(c,j-1)}$, so $(T_j+1)v_{(c,j-1)} = 0$.
- In $v_{(c,j)}$ (where $S = \{c, j\}$, $c < j$): $j \in S$ at $(1, 3)$ (content 2). $j+1 \notin S$ (since $c < j$); column 1 set is $\{2, \dots, n\} \setminus \{c, j\}$, sorted; elements below $j+1$ are $\{2, \dots, j\} \setminus \{c, j\} = \{2, \dots, j-1\} \setminus \{c\}$ ($j-3$ elements). So $j+1$ is at $(j-2, 1)$, content $-(j-3)$.

The T_j -block partner is $v_{s_j(c,j)} = v_{(c,j+1)}$ (swap $j, j+1$ moves $j+1$ to row 1, j to column 1), with axial distance $d = 2 - (-(j-3)) = j-1$.

Wait, this gives $d = j-1$, not $d = j$ as claimed. Let me re-check.

Hmm, the axial distance depends on n and the position of $j+1$ in column 1. Let me recompute carefully.

For $S = \{c, j\}$ with $c < j$: column-1 set sorted is

$$\{2, 3, \dots, c-1, c+1, c+2, \dots, j-1, j+1, j+2, \dots, n\}.$$

(Two omissions: c and j .) Elements strictly below $j+1$: $\{2, \dots, c-1, c+1, \dots, j-1\}$, total $j-3$ elements (since $\{2, \dots, j-1\}$ has $j-2$ elements, minus 1 for c). So $j+1$ is the $(j-2)$ th smallest (rank $j-2$), at cell $(j-1, 1)$, content $-(j-2)$.

Axial distance: $d = (\text{content of cell of } j) - (\text{content of cell of } j+1) = 2 - (-(j-2)) = j$.

I had an off-by-one above; the correct value is $d = j$, not $j-1$. So:

The T_j -block on $\{v_{(c,j)}, v_{(c,j+1)}\}$ has axial distance $d = j$ (for any c with $2 \leq c \leq j-1$).

By the Hoefsmit projection formula, $(T_j+1)v_{(c,j)} = c_j u_{(c,j)}^+$ where $u_{(c,j)}^+ = \alpha_j v_{(c,j)} + \beta_j v_{(c,j+1)}$.

Combining (i):

$$(T_j+1)u_{(c,j-1)}^+ = \alpha_{j-1} \cdot 0 + \beta_{j-1} \cdot c_j u_{(c,j)}^+ = \beta_{j-1} c_j u_{(c,j)}^+.$$

(ii) *The middle vector* $v_{(j-1,j)}$. $j-1$ at $(1, 2)$, j at $(1, 3)$. Both in row 1. T_j couples row-1 pair? Wait T_j swaps j and $j+1$, not $j-1$ and j . Position of $j+1$: column 1 set = $\{2, \dots, n\} \setminus \{j-1, j\}$, sorted: $\{2, \dots, j-2, j+1, \dots, n\}$ (omitting $j-1$ and j). So $j+1$ is the $(j-2)$ th smallest, at cell $(j-1, 1)$, content $-(j-2)$.

Axial distance: $d = 2 - (-(j-2)) = j$ (content of j at $(1, 3)$ is 2).

T_j -block partner: swap j and $j+1$ in $T_{(j-1,j)}$ gives $j+1$ at $(1, 3)$, j at $(j-1, 1)$. New SYT has $S' = \{j-1, j+1\}$, i.e. the SYT $T_{(j-1,j+1)}$.

So $(T_j+1)v_{(j-1,j)} = c_j \cdot u_{(j-1,j)}^+$ where $u_{(j-1,j)}^+ = \alpha_j v_{(j-1,j)} + \beta_j v_{(j-1,j+1)}$. This matches the form $u_{(c,j)}^+$ with $c = j-1$ in our formula (1).

(iii) *The right vectors* $u_{(j-1,b)}^+ = \alpha_{j-1} v_{(j-1,b)} + \beta_{j-1} v_{(j,b)}$ for $j+1 \leq b \leq n$. Two cases:

- Sub-case $b = j+1$: $u_{(j-1,j+1)}^+ = \alpha_{j-1} v_{(j-1,j+1)} + \beta_{j-1} v_{(j,j+1)}$.

$T_j v_{(j-1,j+1)}$: $S = \{j-1, j+1\}$, so $j \notin S$ (in column 1), $j+1 \in S$ at $(1, 3)$. Position of j : column-1 set $\{2, \dots, n\} \setminus \{j-1, j+1\}$, sorted; j is the $(j-2)$ th smallest (elements below it: $\{2, \dots, j-1\} \setminus \{j-1\} = \{2, \dots, j-2\}$, $j-3$ elements; so j is at $(j-1, 1)$, content $-(j-2)$). T_j -block: swap $j, j+1$ moves j to $(1, 3)$, $j+1$ to $(j-1, 1)$; new SYT has $S' = \{j-1, j\}$, i.e. $T_{(j-1,j)}$.

Axial distance from $j+1$'s cell $(1, 3)$ (content 2) to j 's cell $(j-1, 1)$ (content $-(j-2)$): $d = 2 - (-(j-2)) = j$. Hence $(T_j+1)v_{(j-1,j+1)} = c_j \cdot u_{(j-1,j)}^+$ where $u_{(j-1,j)}^+ = \alpha_j v_{(j-1,j)} + \beta_j v_{(j-1,j+1)}$.

$T_j v_{(j,j+1)}$: $S = \{j, j+1\}$, both j and $j+1$ in row 1. $T_j v = qv$, so $(T_j+1)v_{(j,j+1)} = (1+q)v_{(j,j+1)}$.

Combining: $(T_j+1)u_{(j-1,j+1)}^+ = \alpha_{j-1} c_j u_{(j-1,j)}^+ + \beta_{j-1} (1+q)v_{(j,j+1)}$.

- Sub-case $b \geq j+2$: $u_{(j-1,b)}^+ = \alpha_{j-1} v_{(j-1,b)} + \beta_{j-1} v_{(j,b)}$.

$T_j v_{(j-1,b)}$: $j \notin S = \{j-1, b\}$, $j+1 \notin S$ (since $b \geq j+2$). Both in column 1. Same column, adjacent. $(T_j+1)v_{(j-1,b)} = 0$.

$T_j v_{(j,b)}$: $j \in S$ at $(1, 2)$, $j+1 \notin S$ in column 1 at $(j, 1)$, content $-(j-1)$ (using same counting; $j+1$ is the $(j-1)$ th smallest in $\{2, \dots, n\} \setminus \{j, b\}$, yielding cell $(j, 1)$). Axial distance

$d = 1 - (-(j-1)) = j$. T_j -block partner: swap $j, j+1$ gives $S' = \{j+1, b\}$, i.e. $T_{(j+1,b)}$. So $(T_{j+1})v_{(j,b)} = c_j \cdot u_{(j,b)}^+$ where $u_{(j,b)}^+ = \alpha_j v_{(j,b)} + \beta_j v_{(j+1,b)}$.

Combining: $(T_{j+1})u_{(j-1,b)}^+ = \beta_{j-1} c_j u_{(j,b)}^+$ for $b \geq j+2$.

Conclusion of inductive step. Putting (i)–(iii) together:

$$\begin{aligned} (T_{j+1}) W_{j-1}^{(n)} &= \text{span} \left(\{u_{(c,j)}^+ : 2 \leq c \leq j-2\} \right. && \text{from (i)} \\ &\cup \{u_{(j-1,j)}^+\} && \text{from (ii)} \\ &\cup \{c_1 u_{(j-1,j)}^+ + c_2 v_{(j,j+1)}\} && \text{from (iii) sub-case } b = j+1 \\ &\left. \cup \{u_{(j,b)}^+ : j+2 \leq b \leq n\} \right) && \text{from (iii) sub-case } b \geq j+2. \end{aligned}$$

The first two contributions together with the linear combination from the third can be re-expressed (by elementary row operations on the basis) as

$$\{u_{(c,j)}^+ : 2 \leq c \leq j-1\} \cup \{v_{(j,j+1)}\} \cup \{u_{(j,b)}^+ : j+2 \leq b \leq n\},$$

which matches (1). This completes the inductive step. $\square$

Corollary 6 (Image of R'_n on $V_{(3,1^{n-3})}$). $W_n := R'_n V_{(3,1^{n-3})}$ has dimension $n-2$ with basis

$$\{u_{(c,n-1)}^+ : 2 \leq c \leq n-2\} \cup \{v_{(n-1,n)}\}.$$

Proof. Apply Proposition 5 at $j = n-1$. The third set $\{u_{(j,b)}^+ : j+2 \leq b \leq n\}$ becomes $\{u_{(n-1,b)}^+ : n+1 \leq b \leq n\}$, which is empty. This leaves $\{u_{(c,n-1)}^+ : 2 \leq c \leq n-2\} \cup \{v_{(n-1,n)}\}$, dimension $(n-3) + 1 = n-2$. Note: $u_{(c,n-1)}^+ = \alpha v_{(c,n-1)} + \beta v_{(c,n)}$, so W_n is supported on subsets $S = (a, b)$ with $b \in \{n-1, n\}$. $\square$

4 Phase B: collapse to E_1^- via R'_{n-1}

We now apply $R'_{n-1} = (T_{n-2}+1) \cdots (T_1+1)$ to W_n . Write $\widetilde{W}_j$ for the image after the partial product $(T_j+1) \cdots (T_1+1)$ acts on W_n .

Proposition 7 (Step 1: collapse to dim 1). $\widetilde{W}_1 = \text{span}(u_{(2,n-1)}^+)$, 1-dimensional.

Proof. By Corollary 6, W_n has the basis $\{u_{(c,n-1)}^+ : 2 \leq c \leq n-2\} \cup \{v_{(n-1,n)}\}$. For $c \geq 3$: $u_{(c,n-1)}^+ = \alpha v_{(c,n-1)} + \beta v_{(c,n)}$ has supports with $2 \notin S$ (since $S = (c, n-1)$ or (c, n) with $c \geq 3$). By Remark 2, $u_{(c,n-1)}^+ \in E_1^-$, so $(T_1+1)u_{(c,n-1)}^+ = 0$. Similarly $v_{(n-1,n)} \in E_1^-$ for $n-1 \geq 3$ (i.e. $n \geq 4$).

Only $u_{(2,n-1)}^+ = \alpha v_{(2,n-1)} + \beta v_{(2,n)}$ survives, and $\chi(2) = 1$ for both supports, so $u_{(2,n-1)}^+ \in E_1^+$ and $(T_1+1)u_{(2,n-1)}^+ = (1+q)u_{(2,n-1)}^+$. $\square$

Proposition 8 (Steps 2 through $n-3$: shifting window). For $2 \leq j \leq n-3$,

$$\widetilde{W}_j = \text{span}(\widetilde{u}_j), \quad \widetilde{u}_j = A_j v_{(j,n-1)} + B_j v_{(j+1,n-1)} + C_j v_{(j,n)} + D_j v_{(j+1,n)},$$

with $A_j, B_j, C_j, D_j \in \mathbb{Q}(q)$ all nonzero (depending on j, n, q).

Proof. By induction on j .

Base case $j = 2$. Apply (T_2+1) to $\widetilde{W}_1 = \text{span}(u_{(2,n-1)}^+)$. By the analysis in Proposition 5 applied at $i = 2$:

- $T_2v_{(2,n-1)}$: 2 at $(1, 2)$, 3 at $(2, 1)$ (column-1 position). T_2 -block with $v_{(3,n-1)}$, axial distance $d = 2$. $(T_2+1)v_{(2,n-1)} = c_2u^+$ where $u^+ = \alpha_2v_{(2,n-1)} + \beta_2v_{(3,n-1)}$.
- $T_2v_{(2,n)}$: similarly T_2 -block with $v_{(3,n)}$ at $d = 2$. $(T_2+1)v_{(2,n)} = c_2(\alpha_2v_{(2,n)} + \beta_2v_{(3,n)})$.

So $(T_2+1)u_{(2,n-1)}^+ = (1+q)c_2(\alpha\alpha_2v_{(2,n-1)} + \alpha\beta_2v_{(3,n-1)} + \beta\alpha_2v_{(2,n)} + \beta\beta_2v_{(3,n)})$ where $(\alpha, \beta) = (\alpha_{n-1}, \beta_{n-1})$ (the T_{n-1} -block coefficients from $u_{(2,n-1)}^+$). All four coefficients nonzero.

Inductive step. Assume $\widetilde{W}_{j-1}$ is the span of $\widetilde{u}_{j-1}$ supported on $\{(j-1, n-1), (j, n-1), (j-1, n), (j, n)\}$ with all coefficients nonzero.

Apply (T_j+1) . By Lemma 1:

- $v_{(j-1,n-1)}$ and $v_{(j-1,n)}$: $S = (j-1, n-1)$ resp. $(j-1, n)$ contain $j-1$ but not j (since $j \neq n-1, n$ for $j \leq n-3$). In column 1, j is at cell $(j-1, 1)$ and $j+1$ at $(j, 1)$ (same column, adjacent — by counting as in Prop. 5 step (i)). So T_j acts as -1 , and (T_j+1) kills these.
- $v_{(j,n-1)}$ and $v_{(j,n)}$: S contains j at $(1, 2)$ but not $j+1$ (since $j+1 \neq n-1, n$ for $j \leq n-3$). T_j -block: $\{v_{(j,b)}, v_{(j+1,b)}\}$ at axial distance $d = j$ (for $b = n-1$ or n). $(T_j+1)v_{(j,b)} = c_j(\alpha_jv_{(j,b)} + \beta_jv_{(j+1,b)})$.

Hence

$$(T_j+1)\widetilde{u}_{j-1} = c_j(B_{j-1}(\alpha_jv_{(j,n-1)} + \beta_jv_{(j+1,n-1)}) + D_{j-1}(\alpha_jv_{(j,n)} + \beta_jv_{(j+1,n)})),$$

which is a single nonzero vector supported on $\{(j, n-1), (j+1, n-1), (j, n), (j+1, n)\}$ with all coefficients nonzero (since $B_{j-1}, D_{j-1}, \alpha_j, \beta_j \neq 0$). $\square$

Proposition 9 (Step $n-2$: terminal shape). $\widetilde{W}_{n-2}$ is 1-dimensional, supported on the 5 subsets

$$\{(n-3, n-2), (n-3, n-1), (n-2, n-1), (n-2, n), (n-1, n)\},$$

and contained in E_1^- .

Proof. Apply $(T_{n-2}+1)$ to $\widetilde{W}_{n-3}$, which is the span of $\widetilde{u}_{n-3}$ supported on $\{(n-3, n-1), (n-2, n-1), (n-3, n), (n-2, n)\}$.

Compute $(T_{n-2}+1)$ on each support:

- $v_{(n-3,n-1)}$: $S = \{n-3, n-1\}$. $n-2 \notin S$ in column 1; $n-1 \in S$ at $(1, 3)$. Column 1 set sorted: $\{2, \dots, n-4, n-2, n\}$ (omitting $n-3, n-1$). Position of $n-2$: it is the $(n-4)$ th smallest (below it: $\{2, \dots, n-4\}$, $n-5$ elements), at cell $(n-3, 1)$, content $-(n-4)$. T_{n-2} -block: swap $n-2, n-1$ gives new SYT $T_{(n-3,n-2)}$ ($n-1$ moves to column 1, $n-2$ moves to $(1, 3)$). Axial distance $d = 2 - (-(n-4)) = n-2$. Hence $(T_{n-2}+1)v_{(n-3,n-1)} = c \cdot (\alpha v_{(n-3,n-1)} + \beta v_{(n-3,n-2)})$.
- $v_{(n-2,n-1)}$: $S = \{n-2, n-1\}$, both at $(1, 2), (1, 3)$. Same row. $(T_{n-2}+1)v = (1+q)v_{(n-2,n-1)}$.
- $v_{(n-3,n)}$: $S = \{n-3, n\}$. $n-2, n-1 \notin S$, both in column 1. Column 1 set: $\{2, \dots, n-4, n-2, n-1\}$. $n-2$ at $(n-3, 1)$, $n-1$ at $(n-2, 1)$. Same column, adjacent. $(T_{n-2}+1)v = 0$.
- $v_{(n-2,n)}$: $S = \{n-2, n\}$. $n-2$ at $(1, 2)$. $n-1 \notin S$ in column 1: column-1 set $\{2, \dots, n-3, n-1\}$, $n-1$ at last position $(n-2, 1)$, content $-(n-3)$. T_{n-2} -block: swap $n-2, n-1$ gives $T_{(n-1,n)}$. Axial distance $d = 1 - (-(n-3)) = n-2$. $(T_{n-2}+1)v_{(n-2,n)} = c \cdot (\alpha v_{(n-2,n)} + \beta v_{(n-1,n)})$.

So if $\tilde{u}_{n-3} = Av_{(n-3,n-1)} + Bv_{(n-2,n-1)} + Cv_{(n-3,n)} + Dv_{(n-2,n)}$ with $A, B, C, D \neq 0$, then

$$\tilde{u}_{n-2} := (T_{n-2}+1)\tilde{u}_{n-3} = c\alpha A v_{(n-3,n-1)} + c\beta A v_{(n-3,n-2)} + B(1+q)v_{(n-2,n-1)} + 0 + c\alpha D v_{(n-2,n)} + c\beta D v_{(n-1,n)}.$$

Five nonzero coefficients; supports as claimed.

For E_1^- containment: every support has minimum element $\geq n-3$, which is ≥ 3 for $n \geq 6$. Hence $2 \notin S$ for every support, so by Remark 2, $\tilde{u}_{n-2} \in E_1^-$. $\square$

Proof of Theorem 3. Combine Corollary 6 (Phase A) and Propositions 7, 8, 9 (Phase B). $B_{n-1}V_\lambda = R'_{n-1}R'_nV_\lambda = \widetilde{W}_{n-2}$ is one-dimensional, supported on the 5 subsets listed, and contained in E_1^- . $\square$

5 Computational verification

Theorem 10 (Verification at $n = 6, 7, 8$). *Theorem 3 and the intermediate steps (Propositions 5, 7, 8, 9) hold computationally at $n = 6, 7, 8$.*

Verification. SymPy at $q = 7 \in \mathbb{Q}$ (generic for the seminormal basis). At each step, the rank, support pattern, and E_1^- containment match the predicted values. At $n = 8$, $\widetilde{W}_{n-2}$ is supported on $\{(5, 6), (5, 7), (6, 7), (6, 8), (7, 8)\}$, all with min element $5 \geq 3$, hence in E_1^- . Script: `~/projects/scratch/2026-05-12-hook-j-formula/verify_k3-proof.py`. $\square$

6 The Catalan support phenomenon

The structural proof above suggests a much broader phenomenon for arbitrary hooks $\lambda = (k, 1^{n-k})$. Computational evidence at $n \leq 8$ for $k = 2, 3, 4$ supports the following.

Conjecture 11 (Catalan support for hooks). *Let $\lambda = (k, 1^{n-k}) \vdash n$ be a hook in the rank-zero regime ($\ell(\lambda) = n - k + 1 > \lceil n/2 \rceil$, equivalently $n \geq 2k$). Then the unique (up to scalar) spanning vector of $B_{\ell+1}V_\lambda$ has support*

- (1) *contained in subsets of the “top range” $R_k(n) := \{n - 2k + 3, n - 2k + 4, \dots, n\}$ (size $2k - 2$);*
- (2) *of cardinality exactly C_k , the k -th Catalan number.*

Theorem 12 (Computational verification of Conjecture 11). *Conjecture 11 holds for $(k, n) \in \{(2, n) : n \geq 4\} \cup \{(3, n) : n \geq 6\} \cup \{(4, 8)\}$ at $n \leq 8$.*

Verification. Compute $B_{\ell+1}V_\lambda$ in the seminormal basis at $q = 7$, identify the support of any nonzero column. Results in Table 1.

In every case, the support lies within the predicted top range and has cardinality exactly C_k . Script: `~/projects/scratch/2026-05-12-hook-j-formula/extract_support.py`. $\square$

6.1 Examples of the Catalan support

Example 13 ($k = 3, n = 6$). The $\binom{4}{2} = 6$ pairs in $R_3(6) = \{3, 4, 5, 6\}$ are $(3, 4), (3, 5), (3, 6), (4, 5), (4, 6), (5, 6)$. The support of $B_5V_{(3,1,1,1)}$ is the $C_3 = 5$ pairs $\{(3, 4), (3, 5), (4, 5), (4, 6), (5, 6)\}$ — all except the “extreme” pair $(3, 6)$.

k	n	λ	top range $R_k(n)$	$ \text{supp} $	C_k
2	4	$(2, 1, 1)$	$\{3, 4\}$	2	2
2	5	$(2, 1^3)$	$\{4, 5\}$	2	2
2	6	$(2, 1^4)$	$\{5, 6\}$	2	2
2	7	$(2, 1^5)$	$\{6, 7\}$	2	2
3	6	$(3, 1, 1, 1)$	$\{3, 4, 5, 6\}$	5	5
3	7	$(3, 1^4)$	$\{4, 5, 6, 7\}$	5	5
3	8	$(3, 1^5)$	$\{5, 6, 7, 8\}$	5	5
4	8	$(4, 1, 1, 1, 1)$	$\{3, 4, 5, 6, 7, 8\}$	14	14

Table 1: Verification of the Catalan support conjecture.

Example 14 ($k = 4, n = 8$). The $\binom{6}{3} = 20$ triples in $R_4(8) = \{3, \dots, 8\}$ split as $C_4 = 14$ supported and $\binom{2k-2}{k-3} = \binom{6}{1} = 6$ excluded. The 6 excluded triples are

$$\{(3, 4, 7), (3, 4, 8), (3, 5, 8), (3, 6, 8), (3, 7, 8), (4, 7, 8)\}.$$

A combinatorial characterization of the C_k -element family remains to be identified (see Section 7).

Remark 15 (Catalan identity). $C_k = \binom{2k-2}{k-1} - \binom{2k-2}{k-3}$ is a standard form of the Catalan number (e.g. via the ballot problem). For $k = 4$: $\binom{6}{3} - \binom{6}{1} = 20 - 6 = 14 = C_4$. For $k = 5$: $\binom{8}{4} - \binom{8}{2} = 70 - 28 = 42 = C_5$.

6.2 Implications for the j -formula

Corollary 16 (Conditional). *If Conjecture 11 holds for $\lambda = (k, 1^{n-k})$ in the rank-zero regime, then the j -formula (2026-05-11- j -formula.tex, Conj. 1) holds for λ , and moreover the rank-corner formula gives the correct rank ($\kappa(\lambda) = 1$).*

Proof. The top range $R_k(n) = \{n - 2k + 3, \dots, n\}$ has minimum element $n - 2k + 3$. For λ in the rank-zero regime $n \geq 2k$, this minimum is $\geq 3 > 2$, so every supported subset S has $2 \notin S$, hence $v_S \in E_1^-$ by Remark 2. The support has $C_k \neq 0$ elements, hence the spanning vector is nonzero, giving rank exactly 1. $\square$

7 Open questions

1. **Combinatorial characterization of the Catalan support family.** Identify a natural rule (Dyck paths, lattice paths, ballot sequences, etc.) describing which C_k subsets of the top range $R_k(n)$ form the support.
2. **Structural proof of Conjecture 11 for $k \geq 4$.** The argument for $k = 3$ in this paper relies on the image remaining 1-dimensional throughout the second staircase factor R'_{n-1} . For $k \geq 4$, the image of $B_{\ell+1}V_\lambda$ is still 1-dimensional, but intermediate images B_jV_λ for $\ell + 2 \leq j \leq n$ have higher dimension (specifically $\binom{n-1-(n-j+1)}{k-1-(n-j+1)}$). A multi-dimensional shifting argument is needed.
3. **Beyond hooks.** For non-hook λ in the rank-zero regime (e.g. $\lambda = (2, 2, 1^{n-4})$), the Catalan support pattern presumably generalizes, with the support governed by a $\kappa(\lambda)$ -tuple of Catalan-like structures (one per length-preserving corner). Verifying at $n \leq 8$ would be the natural next step.

4. **Closing the rank-zero theorem at $n \geq 8$ even.** The May-11 j -formula (Cor. 1) reduces the rank-zero theorem to the j -formula at boundary $n = 2k$. Conjecture 11 for $k = 4, 5, \dots$ would close the boundary cases successively.

8 Summary

Proven structurally.

1. The j -formula for $\lambda = (3, 1^{n-3})$ at every $n \geq 6$ (Theorem 3).
2. Hence $\Pi^{S_n}|_{V_{(3, 1^{n-3})}} = 0$ unconditionally for every $n \geq 6$ (Corollary 4). Combined with the May-11 result for $(2, 1^{n-2})$, the rank-zero theorem now has a structural proof for two infinite families of hooks.

Computationally verified, structurally open.

1. The Catalan support pattern (Conjecture 11) at $n \leq 8$, $k \leq 4$ (Theorem 12).

Why the $k = 3$ proof works.

Three structural facts conspire:

- After R'_n , the image W_n is the span of $+q$ -eigenvectors of T_{n-1} at axial distance $n-1$, supported on subsets with second element $\in \{n-1, n\}$ (Cor. 6).
- The first factor (T_1+1) of R'_{n-1} collapses W_n to the 1-dim subspace $\text{span}(u_{(2, n-1)}^+)$ (Prop. 7).
- Subsequent factors $(T_2+1), \dots, (T_{n-2}+1)$ shift this 1-dim subspace through the SYT lattice in a controlled way: the *shifting window* of Prop. 8 takes the support from $a \in \{2, 3\}$ to $a \in \{n-3, n-2\}$, eventually landing in the 5-element configuration of Prop. 9.

For $k \geq 4$, the second step does not collapse to 1-dim; instead the image of $(T_1+1) \cdot W_n$ has dimension $\binom{n-3}{k-3} > 1$. A higher-dimensional shifting argument is required.

Honest assessment. This paper closes the rank-zero theorem for one more infinite family ($k = 3$ hooks) and provides strong computational evidence for a Catalan structure governing the kernel of $B_{\ell+1}$ on all hook irreducibles. The structural proofs for $k \geq 4$, the combinatorial characterization of the C_k -family, and the boundary $n = 2k$ cases for the rank-zero theorem all remain open.